GateMate™ FPGA

PHY Interface for the PCIe Architecture

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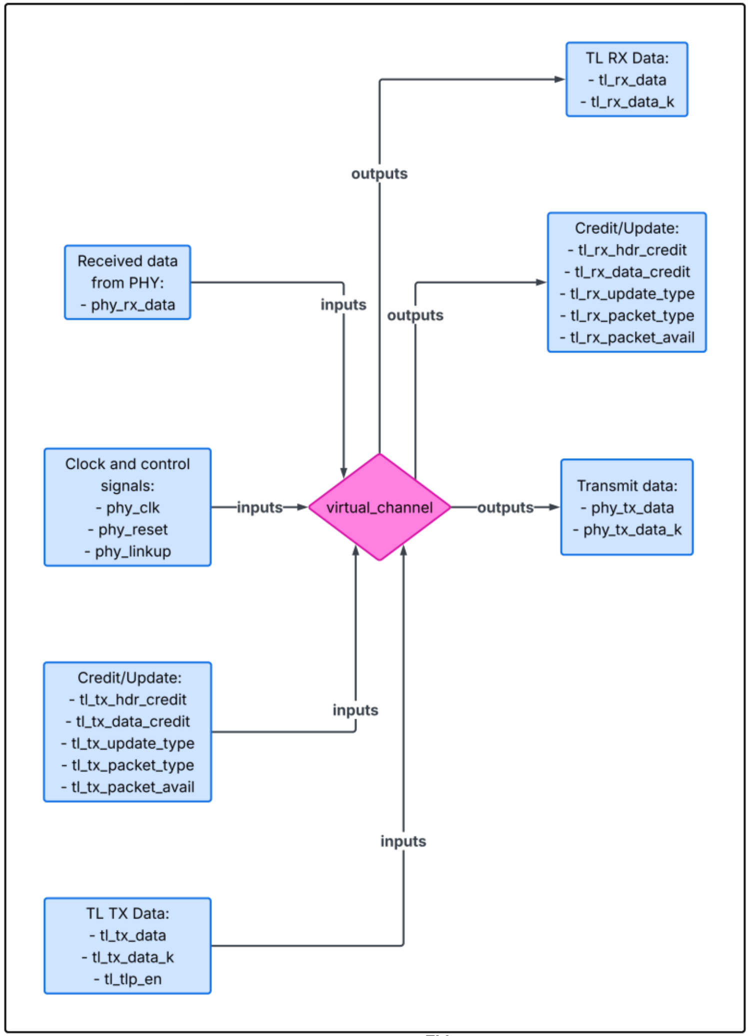
DLL Interface

Virtual Channel Diagram



DLL Port List

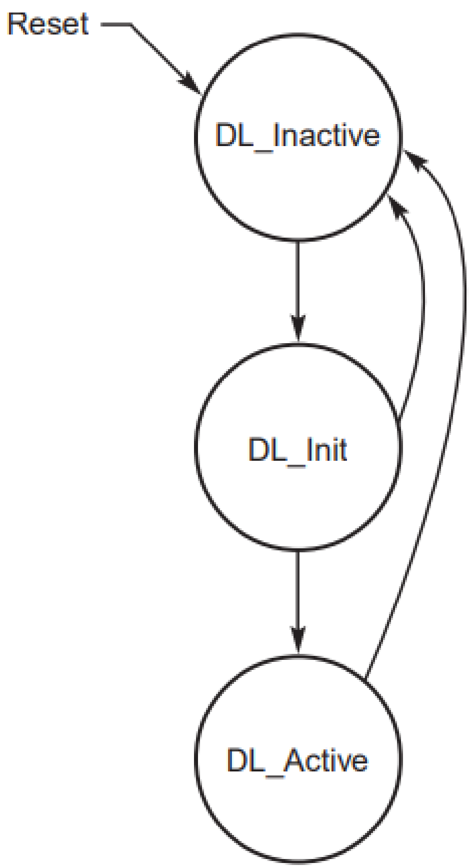
PHY Interface

Direction	Name	Description
Input	phy_clk	PHY-domain clock
Input	phy_reset	Async reset from the physical layer
Input	phy_linkup	Link-up indication from the physical layer
Input	phy_rx_data	Receive data stream from the physical layer
Input	phy_rx_data_k	K-character (control) markers for phy_rx_data
Output	phy_tx_data	Transmit data stream to the physical layer
Output	phy_tx_data_k	K-character (control) markers for phy_tx_data

TL Interface

Direction	Name	Description
Input	tl_tx_data	TLP payload from the transaction layer
Input	tl_tx_data_k	K-character markers for tl_tx_data
Input	tl_tlp_en	Valid flag for tl_tx_data
Input	tl_tx_hdr_credit	TX flow-control header credits from TL
Input	tl_tx_data_credit	TX flow-control data credits from TL
Input	tl_tx_update_type	TX flow-control update type
Input	tl_tx_packet_type	TX packet type
Input	tl_tx_packet_avail	DLLP tx valid
Output	tl_rx_data	Received TLP payload delivered to TL
Output	tl_rx_data_k	K-character markers for tl_rx_data
Output	tl_rx_hdr_credit	RX flow-control header credits reported to TL
Output	tl_rx_data_credit	RX flow-control data credits reported to TL
Output	tl_rx_update_type	RX flow-control update class
Output	tl_rx_packet_type	RX packet class
Output	tl_rx_packet_avail	DDLP rx valid

DLL Init FSM



State Transition Conditions

`Reset → DL_Inactive`

Asserted on power-up or whenever the data link is reset.

`DL_Inactive → DL_Init`

Entered when the physical layer reports the link is up (phy_linkup = 1) and the data link layer is not disabled.

`DL_Init → DL_Active`

Flow-control initialisation (FC_INIT1 and FC_INIT2 exchange) has completed successfully for all virtual channels.

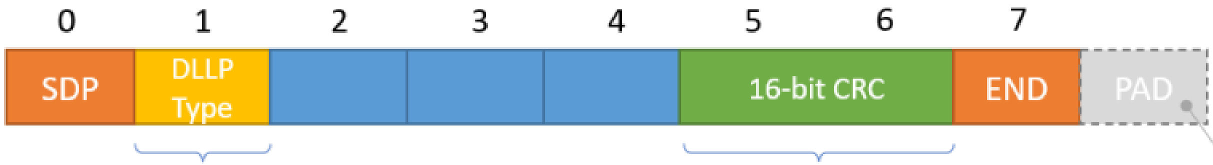
`DL_Active → DL_Inactive`

The physical layer reports loss of link (phy_linkup = 0), a hot reset is received, or the link is disabled.

`DL_Init → DL_Inactive`

Link goes down (phy_linkup = 0) before flow-control initialisation can complete.

DLLP structure



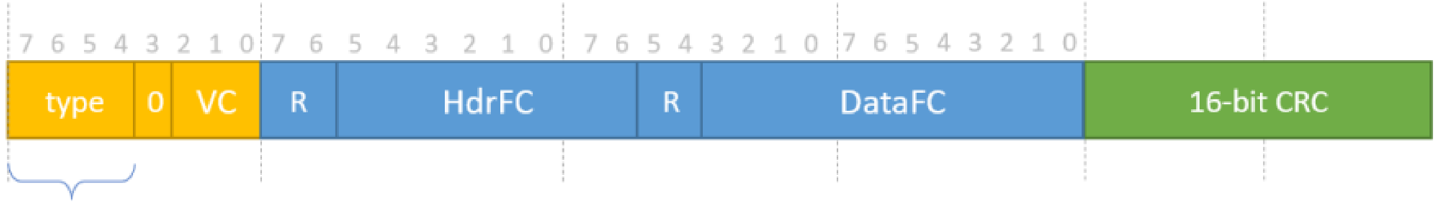
0000 0000: ACK
0001 0000: NAK
0010 0000: PM_Enter_L1
0010 0001: PM_Enter_L23
0010 0011: PM_Active State_Request_L1
0010 0100: PM_Request_Ack
0011 0000: Vendor specific
0100 0xxx: InitFC1-P (xxx = virtual channel #)
0101 0xxx: InitFC1-NP
0110 0xxx: InitFC1-Cpl
1100 0xxx: InitFC2-P
1101 0xxx: InitFC2-NP
1110 0xxx: InitFC2-Cpl
1000 0xxx: UpdateFC-P
1001 0xxx: UpdateFC-NP
1010 0xxx: UpdateFC-Cpl

$$G(x) = x^{16} + x^{12} + x^3 + x + 1$$

Initial value is FFFFh
DLLP bytes fed in LSb first
Bytes of result inverted and bit reversed

PADs to link width if no new DLLP or TLP to transmit

DLLP structure



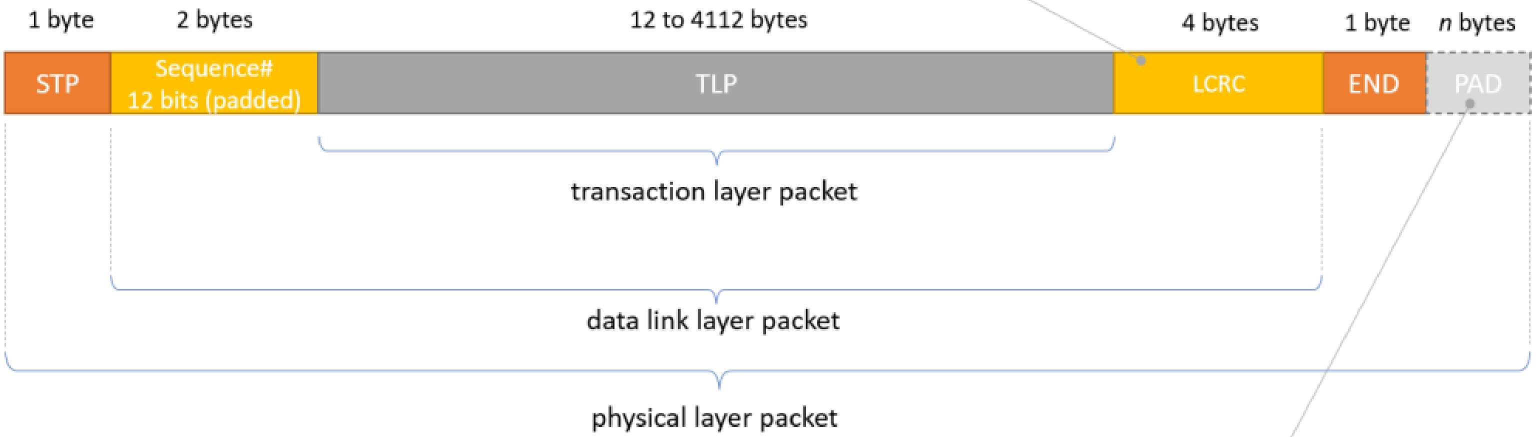
0100b = InitFC1-P
0101b = InitFC1-NP
0110b = InitFC1-Cpl
1100b = InitFC2-P
1101b = InitFC2-NP
1110b = InitFC2-Cpl
1000b = UpdateFC-P
1001b = UpdateFC-NP
1010b = UpdateFC-Cpl

DLLP Type = {tl_tx_packet_type (2 bits), tl_rx_update_type (2 bits), 1’b0, virtual channel number (3 bits)}

TLP structure

$$G(x) = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^{8} + x^{7} + x^{5} + x^{4} + x^{2} + x + 1$$

Initial value is FFFFFFFFh
DLLP bytes fed in LSb first
Bytes of result inverted and bit reversed



PADs to link width if no new DLLP or TLP to transmit

TLP Framing, LCRC & Acknowledgement

Framing and LCRC

- A TLP is bracketed by STP and END tokens, analogous to the SDP/END framing of a DLLP. An additional EDB (end-data-bad) token may replace END to terminate a transaction early on the link.

- A transaction may be aborted without transmitting all of its data provided it is closed with EDB at the PHY and the LCRC bits are inverted relative to a valid packet. This signals the receiver to discard the packet.

- The data link layer prepends a 12-bit sequence number and appends a 32-bit Link CRC (LCRC). As with the DLLP CRC, bytes are fed LSb-first and the result is bit-reversed and inverted before transmission.

- See the polynomial and initial value on the previous slide.

Sequence Numbers and ACK / NAK

- Sequence numbers start at 0 (initialised on DL_LinkDown) and are incremented for each transmitted TLP. The LCRC covers both the sequence number and the TLP payload.

- On reception the LCRC is checked. A valid CRC triggers an ACK DLLP; a failure triggers a NAK DLLP carrying the sequence number of the last correctly received TLP.

- The ACK/NAK DLLP payload is the 12-bit ACK/NAK sequence number; the Type field distinguishes ACK from NAK.

- The transmitter holds every sent TLP in a retry buffer. On ACK, all TLPs with a sequence number at or below the ACK value are released. On NAK, those TLPs are released and the remaining unacknowledged TLPs are retransmitted oldest-first.

TLP Retries, Timeout & Error Reporting

Retry Limit and Link Retraining

- The number of retries is capped at four. A fifth retry attempt causes the data link layer to request that the physical layer retrain the link, while the data link and transaction layers themselves are not reset.

- When link training completes, the data link layer retransmits all unacknowledged TLPs from the retry buffer in their original order (oldest first).

- A replay timer at the transmitter monitors whether any TLP in the retry buffer has been neither ACK'd nor NAK'd. Expiration of this timer initiates a retry. The timeout value is a function of maximum payload size, link width, and other latencies. Timer expiration is a reported error.

Error Reporting and Sequence-Number Rollover

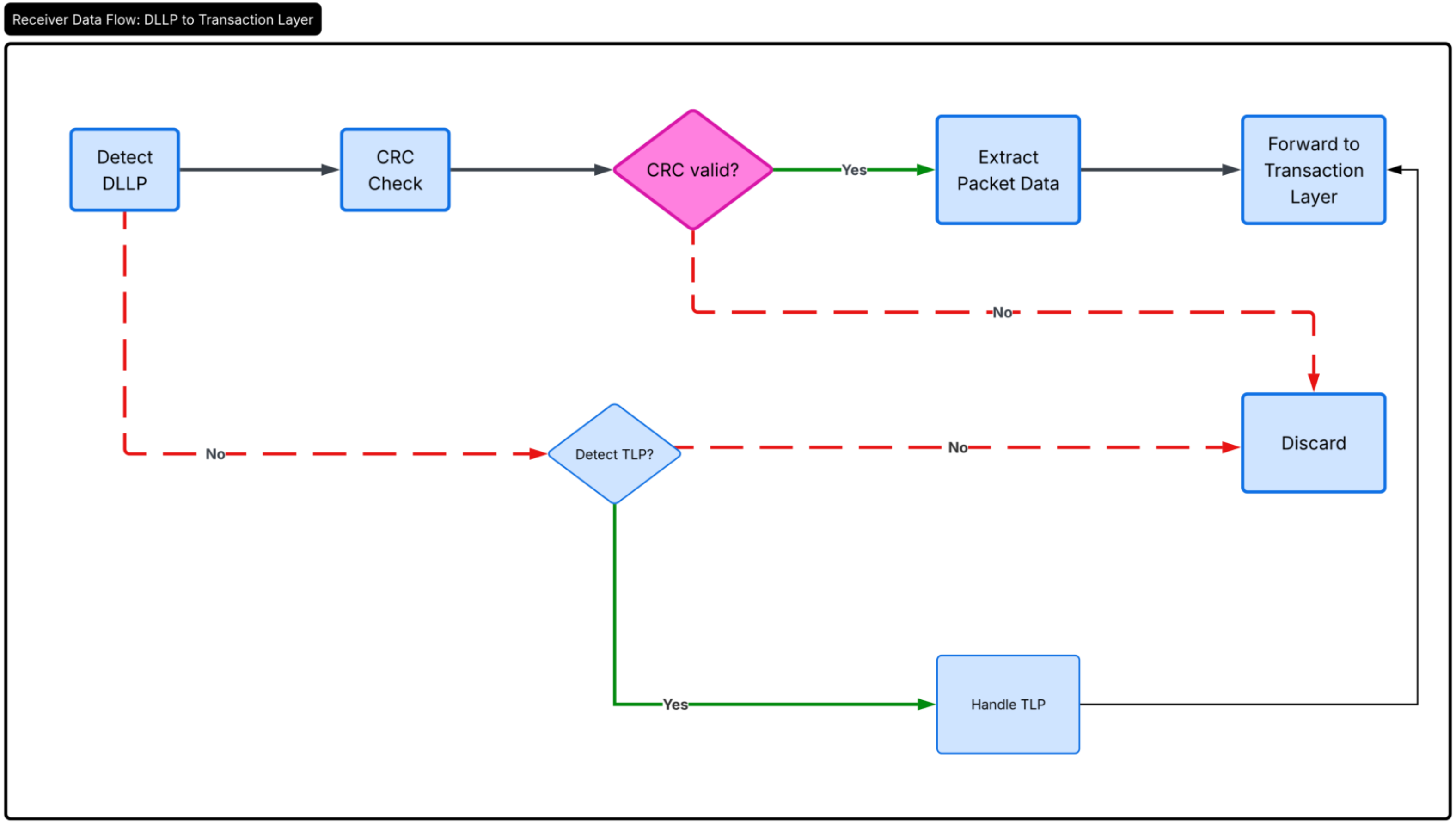
- Reception of a TLP with an unexpected sequence number indicates that one or more TLPs have been lost. This is also a reported error.

- Together, these mechanisms ensure the link does not lock up on transient errors while still surfacing persistent link faults to higher layers through reported errors.

- Sequence numbers are 12 bits wide (0 – 4095). To guarantee unambiguous rollover, and analogously to the flow-control credit scheme, the maximum number of outstanding (unacknowledged) TLPs is limited to half this range — 2048 — regardless of any additional credits available.

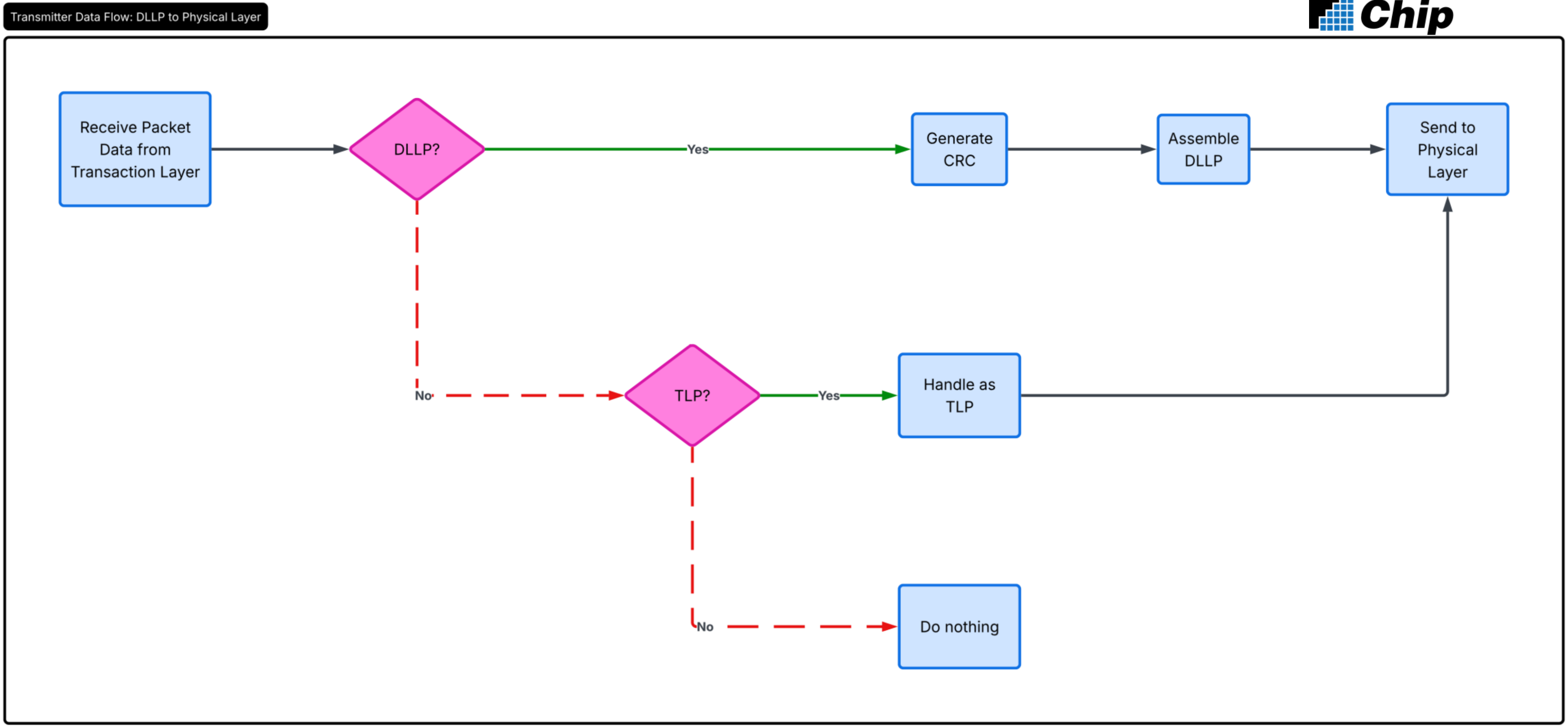
- When this limit is reached, the data link layer stops accepting further TLPs from the transaction layer until ACKs free retry-buffer entries.

RX flow



Also, DLLP Update must be scheduled for sending at regular intervals. This defaults to 30µs (-0%/+50%) but can be configured to be 120µs

TX flow


Transmitter Data Flow: DLLP to Physical Layer
Receive Packet Data from Transaction Layer
DLLP?
Yes
Generate CRC
Assemble DLLP
Send to Physical Layer
No
TLP?
Yes
Handle as TLP
No
Do nothing

Contact

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